

Yejin Jung

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EDUCATION

Arkansas State University

B.S. in Computer Science, GPA: 3.87/4.00

Jonesboro, AR

Aug 2024 – Dec 2026 (Expected)

Sangmyung University

B.S. Coursework in Human-Centered Artificial Intelligence, GPA: 4.29/4.50

Seoul, South Korea

Mar 2022 – Dec 2023

RESEARCH INTERESTS

Machine Learning, Deep Learning, Computer Vision, Video Understanding, Embodied AI

RESEARCH EXPERIENCE

Undergraduate Research Assistant

Aug 2024 – May 2025, Feb 2026 – Present

Arkansas State University

Jonesboro, AR

- Conducted research on antimicrobial resistance prediction using bacterial whole-genome sequencing data.
- Developed and evaluated multiple machine learning and deep learning models for predicting antimicrobial resistance.
- Designed genomic feature extraction, data preprocessing, and model evaluation pipelines.
- Performed statistical analysis, data visualization, and manuscript preparation, contributing to publications at ACM-BCB Companion 2025 and IEEE BIBM 2026.

PUBLICATIONS

Yejin Jung, Chia Lee, and Donghoon Kim. “Multi-Label Antimicrobial Resistance Prediction for Antibiotic Prioritization.” Accepted as a regular paper at the *2026 IEEE International Conference on Bioinformatics and Biomedicine (IEEE BIBM 2026)*, Dallas, TX, USA, Dec 2026. Acceptance rate: 19.48% (480 accepted out of 2,463 submissions).

Yejin Jung*, Namkyeong Kim*, and Donghoon Kim. “Improving Antimicrobial Resistance (AMR) Phenotype Prediction for Unseen Bacteria through Data Augmentation and Machine Learning.” *ACM-BCB Companion Proceedings (ParBio Workshop)*, 2025.

* Equal contribution.

PROJECTS

AI Closet – AI-Based Outfit Recommendation System | Python, CLIP

Aug 2026 – Present

- Developing an AI-based wardrobe application that recommends outfits based on users’ clothing images, weather conditions, and event context.
- Implemented zero-shot clothing image classification using CLIP and developed a recommendation pipeline that combines classified clothing items with real-time weather and event information.

Multi-Label Antimicrobial Resistance Prediction | Python, PyTorch, scikit-learn

Jan 2026 – Present

- Designed a multi-label deep learning framework for predicting antimicrobial resistance to multiple antibiotics from bacterial whole-genome sequencing data.
- Developed and evaluated CNN, MLP, FT-Transformer, ResNet, and tree-based machine learning models using original and augmented datasets.
- Developed an inference pipeline to evaluate model generalization on external bacterial genomes from NCBI.
- Developing a Python backend to integrate trained models for deployment as a web-based AMR prediction application.

Digital Logic Circuit Design & Implementation | *Multisim, Breadboard Circuit Design*

Apr 2026

- Designed and implemented a binary addition/subtraction circuit with 7-segment display output using digital logic ICs, breadboard-based hardware, and Multisim simulation.

AMR Phenotype Prediction using Data Augmentation | *Python, TensorFlow, scikit-learn* Aug 2024 – May 2025

- Developed genomic feature extraction, preprocessing, and data augmentation pipelines for antimicrobial resistance phenotype prediction.
- Trained and evaluated machine learning and deep learning models with hyperparameter optimization, contributing to a publication at ACM-BCB Companion 2025.

AWS & KT AICE Big Data AI Competition | *Python, Computer Vision*

May 2023 – Sep 2023

- Led a student team in developing a computer vision system for analyzing abnormal driving behavior from CCTV footage.
- Applied vehicle detection and tracking to extract and analyze vehicle trajectories for abnormal driving behavior detection.

PRESENTATIONS

“Machine Learning-Based Prioritization of Antibiotic Susceptibility Testing Using Whole-Genome Data,”
CREATE@State Symposium, Arkansas State University, Apr 2026

“Using Machine Learning to Predict Antimicrobial Resistance (AMR) Phenotypes,” CREATE@State Symposium,
Arkansas State University, Apr 2025

“Using Machine Learning to Predict Antimicrobial Resistance (AMR) Phenotypes,” STEM Posters at the Capitol, Little
Rock, Arkansas, Jan 2025

AWARDS & HONORS

Chancellor’s List, Arkansas State University (Spring 2025, Fall 2025, Spring 2026)

3rd Place Poster Award, CREATE@State Symposium, Arkansas State University, Apr 2025

The 1st Korea-U.S. Student Exchange Program in the Field of High-Tech Industry, Jun 2024

Grand Prize, SW-Centered University Generative AI Utilization Contest, Sangmyung University, Nov 2023

Top 7 Finalist, Big Data AI Competition with AWS and KT AICE, Sep 2023

Academic Achievement Award, Sangmyung University, Aug 2023

SW Scholarship (Excellence in Department; ranked 2nd in department), Sangmyung University, Feb 2023

SW Scholarship (Top in Department), Sangmyung University, Aug 2022

TECHNICAL SKILLS

Programming Languages: Python, Java, C/C++, SQL, Kotlin

Machine Learning & Data Science: scikit-learn, Pandas, NumPy, Matplotlib

Deep Learning: PyTorch, TensorFlow

Tools & Platforms: Git, MySQL